

UCE511: TRANSPORTATION ENGINEERING-I				
	L	T	P	Cr
	3	0	2*	3.5
Course Objective: The objective of the course is to enable students study different types of pavements, to analyze road pavement structures, to differentiate between the different types of materials used and to design and construct road pavements. Another objective is to expose the students to geometric design, both vertical and horizontal and to enable the civil engineering students to study the road user characteristics and formulate fundamental principles of traffic flow, traffic characteristic measurements				
General: Different modes of transport, Development of Transport System, Phased development of Roads in India, Planning & Management of Highways, Various road plans developed in India, Road patterns, Highway Surveys & Alignment, Design, Drawings, Estimates & Project Report				
Traffic Studies: Road user characteristics, Importance of traffic studies, spot speed, speed and delay and origin and destination studies. Traffic accident studies, Causes of accidents and Remedial Measures, Parking studies				
Geometric Design of Highways: Introduction, Highways Classification, Right of way, Width of formation, Sight Distances, Stopping site distance, overtaking sight distance, overtaking zones, camber, Road Curves, Transition Curves, Super elevation, Widening at curves, IRC codal recommendations for various geometric design parameters, Road Safety Audits, Highway capacity & Intersection design				
Construction of Roads: Various types of bituminous layers constructions and their selection, specifications for embankments, subgrade, granular subbase, water bound macadam, wet mix macadam, surface dressing, premix carpet, bituminous macadam, dense bituminous macadam, bituminous concrete, mastic asphalt, stone matrix asphalt, dry lean concrete, cement concrete pavements, Importance & Principles of Highway Drainage, Surface Drainage & Sub-Surface drainage				
Types of bituminous binders and Mix design: Manufacturing of bitumen, Paving bitumen specifications as per IS 73: 2013, comparison between bitumen, tar, cut back & emulsion, Modified binders and its rheology, Design of bituminous mixes: Requirement of bitumen mixes, design of bituminous mixes as per Marshall Stability & flow method, parametric evaluation of bituminous mixes, IRC & MORTH recommendations for the design mix of various layers of pavements				
Pavement Design: Factors affecting design of pavements, design principles & design procedures as per IRC 37 guidelines, Design of PQC pavements as per IRC 58 & SP62, Use of software's IITPAVE, KENPAVE				
Failures of flexible and rigid pavements & Highway Maintenance: Causes of Failures and Remedial Measures, Maintenance of flexible and rigid pavements, pavement evaluation and				

its strengthening method, overlay design using IRC 81 & IRC115 guidelines, concept of thin & ultrathin white toppings, design & scheduling of maintenances activities, development of Pavement maintenance management systems for different categories of roads, Economic Evaluation of Highway Projects and Life Cycle Costing Concept

Laboratory Work:

The students will perform various quality control tests as per Indian Road Congress (IRC) & Ministry of Road Transport & Highways (MORTH) specifications for the various layers of the pavement section. Paving bitumen & bituminous mix testing like penetration value, softening point, viscosity & binder rheology, ductility value, centrifugal extraction, aggregate strength tests, pavement layer gradation & stability - flow analysis, Deflection studies for the granular layers of the flexible & rigid pavements

Experimental Project/assignment/Micro Project: Students in groups of 4 to 6 will do the projects

1. Calculate the un-soaked and soaked CBR value of different soil samples
2. Design the flexible pavement as per IRC 37 guidelines for the given traffic data
3. Design the rigid pavements as per the IRC 58 guidelines for the given traffic data

Course Learning Objectives (CLO)

Upon the completion of this course, the students will be able to:

1. Quantify the specifications of various road construction materials required
2. Perform geometric design of highways and expressways
3. Perform analysis and design of flexible and rigid pavements
4. Address highway maintenance, drainage and economic issues
5. Perform the traffic studies necessary before making changes to or designing new road Infrastructure

Text Books:

1. Khanna S.K. and C.E.G. Justo, Highway Engineering, Nem chand Bros (2017)
2. Kadyali L. R.; Highway Engineering, Nem Chand & Brothers, Roorkee (2018)

Reference Books:

1. Sharma & Sharma; Principle and Practice of Highway Engineering, Asia Publishing House, New Delhi (2010).
2. G.V.Rao, Tata McGraw Hill, New Delhi Principles of Transportation and Highway Engineering (2002)
3. Yoder E. J.; Principles of Pavement Design, John Wiley & Sons (2011)

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	25
2.	EST	35
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	40